Joint predictions and complementary measurements

I. Explicit model, fitted inputs and independent predictions

Orientation and shape coordinates; quasiparticle space; conserved quantum numbers

Potential, inertia, pairing and effective operators, with stated assumptions

Intrinsic trajectories and basis components remain model representations.

II. Correlated observables and data held out from fitting

Energy sequence
Crossings and staggering

Lifetimes and rates
Branching, ICC and shared errors

Angular response
Mixed amplitudes and phase conventions

Spin dependence
Alignment and configuration change

Joint response and covariance are needed; derived quantities are not independent experiments.

III. Concrete transitions test specified model implementations

^{98}Mo: weak $E3$ candidates at 1833.3 and 1585.8 keV
Lifetime plus complete same-mother branches tests enhancement versus a weak denominator.

^{102}Ru: separate $E1$ and $M1$ responses
NRF branching and two-phonon decay strengths constrain different interpretations.

^{135}Pr: the 747.3-keV $17/2^- \rightarrow 15/2^-$ link
Compare absolute electromagnetic predictions, root branches and model-input uncertainty.

Conceptual map; arrows do not encode causal proof, measured rates or statistical significance.